Indian Institute of Technology Kanpur
Department of Mathematics and Statistics
Analysis I (MTH 301)
Quiz 3 solutions

Date: 22/10/2024

Time: 40 minutes | Total marks: 12

Comments

1. Question 1 was discussed in the class while proving the equivalent characterizations of compactness before the mid-semester examination.

2. Question 2 was discussed in the class of 03 October 2024, Thursday.

Solutions

1. (a) Define *totally boundedness*.

Answer. *Let (X,d) be a metric space and $A \subseteqq X$. We say that A is totally bounded if for any $\varepsilon > 0$, there exist $n \in \mathbb{N}$ and $x_1, \ldots, x_n \in X$ such that $A \subseteqq \bigcup_{i=1}^{n} B(x_i;\varepsilon)$.*

Common mistakes.

1. *Quantifiers must be written in the correct order. This cannot be compromised.*
2. *It has to be clearly mentioned in the definition that "**for every** $\varepsilon > 0$, **there exist** finitely many open balls with radius ε ..." Quite a few students missed that crucial point. It is not clear from their answers whether the statement is being made for some ε or for all ε.*
3. *A need NOT be equal to $\bigcup_{i=1}^{n} B(x_i;\varepsilon)$. To see this, take A to be any singleton in $\mathbb{R}$.*
4. *Some students tried to be smart and wrote that,* "A is said to be totally bounded if every sequence in A has a convergent subsequence." *This is certainly NOT the definition of totally boundedness.*
5. *Some of you have assumed that A is finite.*

(b) Let X be a metric space $A \subseteqq X$. Show that if A is not totally bounded, then there exists a sequence in A which does not have a convergent subseqeucne.

Solution. *Since X is not totally bounded, there exists $\varepsilon > 0$ such that, for all finite $F \subseteqq X$, one has $\bigcup_{x\in F} B(x;\varepsilon) \subsetneqq X$. We define a sequence $\{x_n\}_{n=1}^{\infty}$ inductively as follows. Let $x_1 \in X$. Assume that we have chosen $x_1, \ldots, x_n \in X$, where $n \in \mathbb{N}$, such that $B(x_i;\varepsilon) \cap B(x_j;\varepsilon) = \varnothing$, whenever $1 \leqslant i \neq j \leqslant n$. Since $\bigcup_{i=1}^{n} B(x_i;\varepsilon) \subsetneqq X$, there exists a point $y \in X$ such that $y \notin B(x_i;\varepsilon)$, for all $i = 1, \ldots, n$. Set $x_{n+1} \overset{\text{def}}{=} y$. By construction, it is clear that,*

$$n < m \Longrightarrow d(x_n, x_m) \geqslant \varepsilon. \tag{1.1}$$

If possible let $\{x_n\}_{n=1}^{\infty}$ admit a convergent subsequence, say $\{x_{n_k}\}_{k=1}^{\infty}$, and let $x_{n_k} \xrightarrow[k\to\infty]{} x$. Then there exist $k \neq k'$ such that $x_{n_k}, x_{n_{k'}} \in B\left(x;\frac{\varepsilon}{4}\right)$. It follows that $d\left(x_{n_k}, x_{n_{k'}}\right) \leqslant \frac{\varepsilon}{2}$, which contradicts (1.1).

Common mistakes.

1. *Please read the following excerpt from item no. 9 of* Guidelines for writing a solution*:*

   "Questions that are directly asked from the lectures and tutorials, exercise sheets are to be answered with all details."

   Therefore the reasoning that if A is not totally bounded then it is not compact, and hence it has a subsequence which does not have a convergent subsequence is NOT acceptable, as this would be a circular argument.
2. *It is NOT true that if A is not totally bounded then it cannot be written as a union of finitely many balls. For instance, $\mathbb{N}$ quipped with the discrete metric if not totally bounded; however it is an open ball with radius 2. this also shows that A can have a finite open cover despite being not totally bounded.*
3. *Some students have written an incorrect definition of totally boundedness. As a consequence, when they tried to prove part (b) by contradiction or otherwise, that too turns out to be complete incorrect.*
4. *A few students have written that if A is not totally bounded then it will be a disjoint union of open balls. This need NOT be true. For example, one can take $\mathbb{R}$ itself.*

[2+3=5]

2. (a) Define *locally constant maps*.

Answer. *Let X, Y be metric spaces and $f : X \longrightarrow Y$. We say that f is* locally constant *if for every $x \in X$, there exists $r > 0$, such that f is constant on $B(x;\varepsilon)$, i.e., $\forall y, z \in B(x;\varepsilon)$, $f(y) = f(z)$.*

Common mistakes.

1. *The quantifiers "for all" and "there exists" need to be in the correct order.*
2. *Some students did not mention whether f is constant on $B(x;r)$ for some x or for all $x \in X$.*
3. *Some answers say "for every $x \in X$ there exists $A \subseteq X$ such that $x \in A$ and f is constant on A". This is NOT correct. It has to be menitioned that A is open.*
4. *There is no meaning of the derivative of a function between two arbitrary metric spaces.*
5. *For $x \in X$, $(x-\varepsilon, x+\varepsilon)$, or any interval makes no sense in a general metric space.*
6. *A locally constant function need not be constant.*
7. *It need not be the case that $X = U_1 \cup U_2 \cup U_3 \cup \ldots$, such that such that each U_i is open, and U_i's are disjoint.*
8. *Y is an arbitrary metric space, need not be $\{0,1\}$.*

(b) Let X and Y be metric spaces and $f : X \longrightarrow Y$ be a locally constant map. Show that, if X is connected, then f must be constant.

Solution. *Fix $x_0 \in X$. Consider $A \stackrel{\text{def}}{=} \{x \in X : f(x) = f(x_0)\}$. Clearly $x_0 \in A$ so that $A \neq \varnothing$.*

Let $x \in A$. Since f is locally constant, there exists $r > 0$ such that $\forall y \in B(x;r), f(y) = f(x)$. It follows that, $\forall y \in B(x;\varepsilon), f(y) = f(x_0)$, i.e., $B(x;\varepsilon) \subseteqq A$. Hence A is open.

Let $z \in X$ be a limit point of A. Since f is locally constant, there exists $\rho > 0$ such that $\forall t \in B(z;\rho)$, $f(t) = f(z)$. As z is a limit point of A, there exists $x \in A \cap B(z;\rho)$. It follows that $f(x_0) = f(x) = f(z)$, i.e., $z \in A$. Hence A is closed. Now from conenctedness, we conclude that $A = X$. Thus f is constant.

Common mistakes.

1. *An open all need not be connected always.*
2. *Y can be an metric space, not necessarily $\{0,1\}$.*
3. *Intervals do not make any sense in an arbitrary metric space.*
4. *If X is not connected then there exist nonempoty disjoint open subsets A and B such that $X = A \cup B$. But there is absolutely no guaranty that $f|_A$ or $f|_B$ will be constant.*

[1+3=4]

3. Let $X \stackrel{\text{def}}{=} \{(x,y) \in \mathbb{C}^2 : xy = 1\}$ be equipped with the standard metric. Is X a path connected metric space? Justify your answer.

Solution. *Consider the map $f : \mathbb{C}\backslash\{0\} \longrightarrow X$, $f(x) \stackrel{\text{def}}{=} \left(x, \frac{1}{x}\right)$. Clearly f is continuous and onto. As $\mathbb{C}\backslash\{0\}$ is path connected, so is X.*

Common mistakes.

1. *Many students have sketched the set $X \stackrel{\text{def}}{=} \{(x,y) \in \mathbb{C}^2 : xy = 1\}$ as an hyperbola. This is NOT correct as $x, y \in \mathbb{C}\backslash\{0\}$. In fact, it is not even a subset of $\mathbb{R}^2$.*
2. *It is true that $X = \{(a,b,c,d) \in \mathbb{R}^4 : ac - bd = 1 \text{ and } ad + bc = 0\}$. From this, it follows that X is the INTERSECTION of the following two subsets:*

$$\{(a,b,c,d) \in \mathbb{R}^4 : ac - bd = 1\} \text{ and } \{(a,b,c,d) \in \mathbb{R}^4 : ad + bc = 0\},$$

NOT their union.

3. *Some students argues as follows: the function $f(z) \stackrel{\text{def}}{=} \frac{1}{z}$ is not defined at 0, hence it is NOT continuous at 0. Consequently, X is not path connected. This is certainly wrong (see the definition of continuity again). No credit has been awarded in such cases.*
4. *$f : \mathbb{C}\backslash\{0\} \longrightarrow \mathbb{C}$, $f(x) \stackrel{\text{def}}{=} \frac{1}{x}$ is continuous, but that does not show the path connectedness of X.*
5. *For $f : X \longrightarrow Y$, where X and Y are metric spaces, $f(X)$ can be connected even if f is not continuous.*
6. *A path in X is a map from $[0,1]$ to X, not from X to $[0,1]$.*
7. *Some of you have considered the path $t \mapsto (1-t)(x_1,y_1)+t(x_2,y_2)$, where $(x_1,y_1), (x_2,y_2)$ are points in X. This is NOT a path in X, i.e., $(1-t)(x_1,y_1) + t(x_2,y_2)$ need not be in X for a generic t in $[0,1]$.*

8. *If* $(x_1, x_2) \in X$, *then* $(1-t)x_1 + tx_2$ *can be* 0 *for some* t, *as* x_1 *and* x_2 *are nonzero COMPLEX numbers. For instance,* $(i, -i) \in X$ *and* $\frac{1}{2}i + \frac{1}{2}(-i) = 0$. *So*

$$\left((1-t)x_1 + tx_2, \frac{1}{(1-t)x_1 + tx_2}\right)$$

may not make sense always.

[3]